

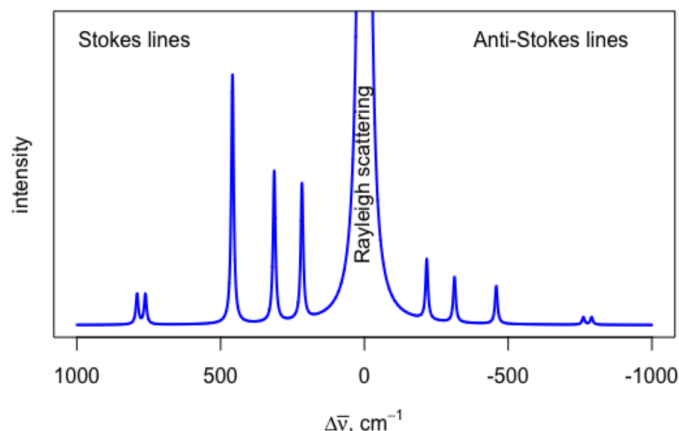
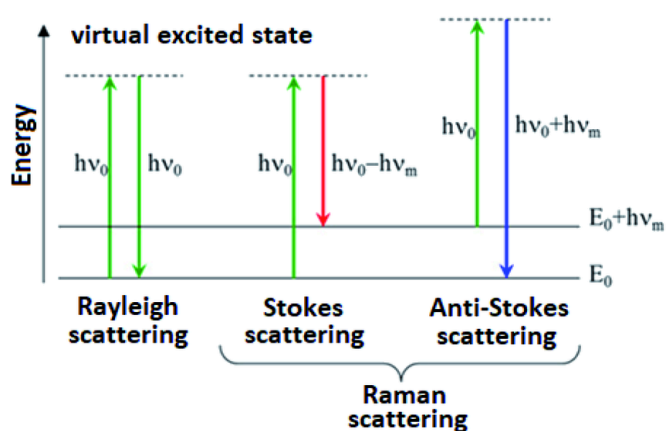
Raman Spectroscopy

What is Raman spectroscopy?

Raman spectroscopy measures the *vibrational modes of molecules* by measuring the inelastic scattering of light caused by a molecule.

How does Raman spectroscopy work?

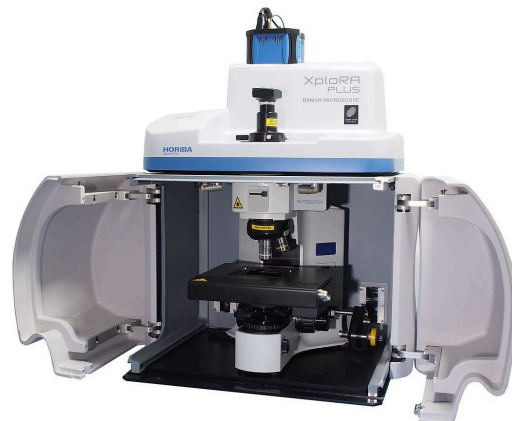
1. A laser in the visible range is shot at your sample, exciting molecules up to a virtual vibrational state.
2. As each molecule relaxes, this can result in one of three outputs.
 - a. **Rayleigh scattering**: a molecule returns the same energy level it originated from, such that $\nu_{\text{Laser}} = \nu_{\text{Rayleigh}}$. This happens the majority of the time.
 - b. **Stokes scattering**: a molecule excited from the ground state and relaxes to a higher vibrational state, such that $\nu_{\text{Laser}} > \nu_{\text{Stokes}}$
 - c. **Anti-stokes scattering**: a molecule excited from a higher energy state and relaxes to the ground state, such that $\nu_{\text{Laser}} < \nu_{\text{Anti-stokes}}$.
3. The scattering of light is measured. The raw data would look like the spectrum below. In reality though, a filter removes Rayleigh scattering (because its intensity is high enough to burn out the detector) and anti-stokes scattering.



What can Raman do that FTIR can't? IR needs a dipole moment. Raman needs a change in polarizability. Thus, Raman uniquely can characterize symmetrical vibrations, given that polarizability (i.e. through volume) changes.	How do you deal with fluorescence? Fluorescence occurs when electrons are pushed up to a higher electronic state. Thus, you need a laser with an energy lower than the difference between energy levels.	Why not use the weakest laser then? Intensity decreases exponentially with the wavelength of the laser. $I \propto \frac{1}{\lambda^4}$ The 4th power!
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Instructions on using the Breakerspace Raman Spectrometer

The Breakerspace has a Horiba XploRA Confocal Raman Microscope, which is capable of collecting Raman spectra using 532 nm, 638 nm and 785 nm lasers.



Sample Preparation:

- The instrument can be used to measure liquids, solids, powders, and gases
- Generally good for organics, polymers, acids/bases, metal oxides, and semiconductors.
- Place sample on a glass microscope slide
- The working distance of the microscope objectives is quite close, so powders should be finely ground and level/smooth
- Solid samples without parallel faces may be mounted to the slide using the sample press and Plastilina mounting clay for a flat surface.
- Prepare your sample externally at the sample prep table

Instrument startup:

- Log in with your MIT Kerberos and open LabSpec 6 software
- Verify instrument is powered on (it should always be left powered on)
- Turn laser emission remote control power to on, if needed

Operation:

- Prepare sample externally on glass slide
- Lower microscope stage using coarse focus knob
- Load slide into slide holder on stage
- Turn brightfield light source brightness to 100%
- Find approximate focus using the coarse knob, adjusting brightness as needed
- Refine focus using joystick rotation
- Close doors on microscope enclosure
- Perform Raman spectroscopy as appropriate for your sample
- Repeat steps as needed.

Instrument shut down:

- Remove sample from microscope stage and close doors
- Turn laser emission remote control power to off
- Save all data (each spectrum must be individually selected and saved)
- Close LabSpec 6 and log off workstation

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!